

High-Performance Filaments with HDT (2026 Reference)

Filament Material Guide

Note: The values maybe different depending on the brands - Check

												C	C	C	C	C	Hr	Un Annealed	Annealed	E
Filament Name	Build Sheet	Cooling Fan Speed	Tg	Nozzle Temp	Bed Temp	Chamber Temp	Drying Temp	Drying Time	HDT at 0.45 Mpa	Annealed HDT at 0.45 Mpa2	Bending Modulus MPa									
ABS	PEI/Glass	Low	105	225	110	60	75	4	95		2300									
ASA	PEI/Glass	Low	105	245	110	60	80	3	95		2000									
CPE	PEI	Medium	90	250	80	20	60	3	70		1900									
HIPS	PEI	0	100	225	100	50	-	-	80		1600									
PA 6	G10/FR4	Low	105	265	110	70	90	4	140		2050									
PA ePA CF	G10/FR4	Low	60	260	90	60	80	10	180		6500									
PA12 CF	G10/FR4	Low	60	260	80	55	80	8	135		3000									
PA12 GF	G10/FR4	Low	60	260	90	50	80	6	150		2900									
PA6 GF	G10/FR4	Low	70	260	90	60	80	10	185		4800									
PC	PEI	0	147	285	115	60	120	4	135		2200									
PC + ABS		Low	137	275	110	80	110	4	126		2370									
PC + ASA		Low	126	265	110	90	110	4	126		2370									
PC BambuLab	PEI	Low	145	260	90	60	80	8	112		2300									
PC Blend CF Prusa	PEI	Low	113	285	110	60	80	5	114	130	4200									
PC BLEND LCF*		Low	158	280	100	90	80	6	185		9800									
PC CF/GF		Low	143	280	110	100	80	5	135		6000									
PC PBT CF		Low	140	260	100	100	80	5	110		6200									
PC Polymax	PEI	Low	113	250	90	70	75	12	114		2040									
PEEK	CF/PEI	Low	143	390	135	100	120	4	140		3700									
PEKK	CF/PEI	Low	165	370	130	150	120	4	182		2900									
PEKK CF	CF/PEI	Low	165	380	120	140	120	6	280		9850									
PET CF	PEI	Low	70	260	70	55	80	10	180		6500									
PETG	PEI	Medium	80	245	70	35	65	3	70		1600									
PLA	PEI	High	85	210	23	25	65	3	80		3500									
PPA CF	PEI	Low	90	300	90	60	100	8	227		9800									
PPS	PEI	0	283	340	120	70	110	4	90		2540									
PPS CF	PEI	0	100	320	110	80	100	10	264		7160									
PPSU	PC/PEI	0	230	400	140	100	120	4	190		2215									
PSU	FR4	0	187	365	130	130	120	4	172		2050									
PVDF		Low	170	265	105	100	-	-	158		2100									
TPC FLEX		Medium	80	235	70	35	65	4	120		100									
ULTEM 1010	PC/PEI	0	186	375	135	110	120	4	208		3000									
ULTEM 9085		0	180	360	130	170	110	5	177		2200									

*Add North

NOTE:

Abrasive Protection:

Glass Fiber (GF) and Carbon Fiber (CF) will destroy brass nozzles. A hardened steel or tungsten carbide nozzle is mandatory

Thermal Treatment:

Semi-crystalline polymers like PPS-CF and PPA-CF often require annealing (baking at ~140°C) to reach their maximum rated HDT

Active Drying:

These materials must be dried at 80°C–120°C for 8+ hours before printing to prevent structural defects